NBA Shot Quality Capstone

Project Plan

DAT 490 — Data Science Capstone (2026 Summer A)

Team Members

Project Lead / Lit Review & Ethics	Cole Campbell
Modeling Lead	Marc Rajesh
EDA & Visualization Lead	Calder Wyllie
Data Engineer	Germain Meza

Executive Summary

This project models the probability that an NBA shot is made based on the context around it — where it was taken, how much time was on the clock, how close the defender was, who took it, and how the game was going. Our data covers every field goal attempt from the 2013-14 season through the current season, roughly two million shots.

From that model we plan to deliver three things: a ranked list of players and teams who consistently outperform what their shots predict, a longitudinal view of how shot success by court location has changed across the tracking era, and a comparison of shot-quality-adjusted player rankings against traditional efficiency metrics.

Profile of the Organization

The National Basketball Association (NBA) is the premier professional men's basketball league in North America and one of the most globally recognized sports leagues in the world. It is the company at the center of our analysis and the stakeholder our findings serve; the league office, individual franchises, and broadcast partners would all benefit from a sharper understanding of shot quality and player evaluation.

Organization Details

Founded: June 6, 1946 (as the Basketball Association of America); merged with the National Basketball League in 1949 to form the NBA as it exists today.

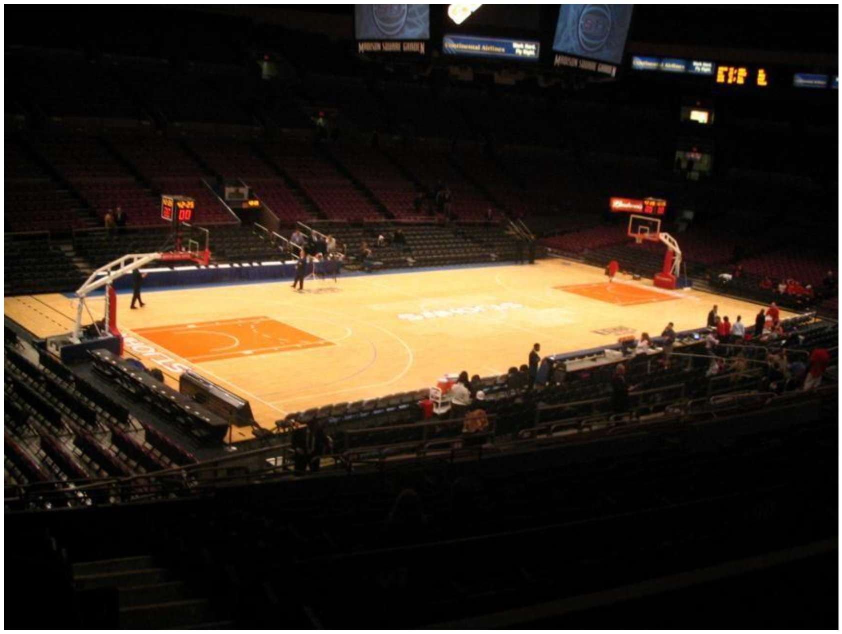
Headquarters: Olympic Tower, 645 Fifth Avenue, New York, NY 10022, United States.

Structure: 30 franchises, divided into the Eastern and Western Conferences. The league operates as a single business entity (NBA, LLC) with individually owned teams. Website: www.nba.com

Business Description

The NBA's core product is professional basketball: 30 teams play an 82-game regular season followed by a 16-team postseason that ends in the NBA Finals. Revenue comes from national and regional media rights, ticket sales, sponsorships, merchandising, and growing international business. The league also operates the WNBA, the NBA G League (its developmental league), the Basketball Africa League, and the NBA 2K League (esports).

Beyond the on-court product, the NBA invests heavily in analytics and player tracking. The league installed optical tracking cameras in every arena starting in the 2013-14 season; that decade of shot-by-shot tracking data is the foundation our analysis builds on.



Madison Square Garden — home of the New York Knicks. Photo: Ganley894, public domain via Wikimedia Commons.

Financials

League revenue: approximately $11 billion annually (2023-24 season) and growing as media-rights deals expand.

Media rights: in 2024 the league signed a new 11-year, ~$76 billion media-rights agreement with Disney (ABC/ESPN), NBCUniversal, and Amazon Prime Video, which kicked off with the 2025-26 season.

Average franchise valuation: approximately $4.4 billion as of 2024 (Forbes annual valuations), with top franchises like the Warriors, Knicks, and Lakers valued above $7 billion each.

Employees: about 1,500 at the league office; teams employ roughly 200-400 staff each, putting total league-wide employment in the range of 10,000-12,000.

Key Executives

Adam Silver — Commissioner (since February 2014)

Mark Tatum — Deputy Commissioner & Chief Operating Officer

Byron Spruell — President, League Operations

Tammy Henault — Chief Marketing Officer

Each of the 30 teams also has its own president of basketball operations, general manager, and analytics staff — the people most likely to use the kind of analysis we're proposing.

Major Competitors

As a professional sports league, the NBA competes for fan attention, broadcast dollars, and athletic talent. The most direct competitors are:

- National Football League (NFL) — the dominant US sports league by revenue and viewership
- Major League Baseball (MLB)

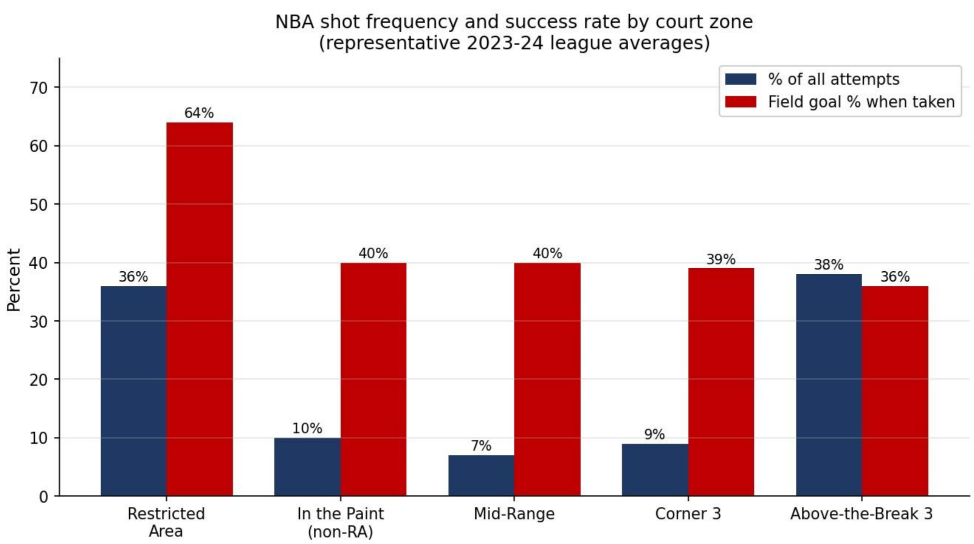
- National Hockey League (NHL)
- Major League Soccer (MLS) — growing quickly in the US market
- International basketball leagues like the EuroLeague and FIBA-sanctioned competitions, especially for talent acquisition
- College basketball (NCAA), especially during March Madness
- Streaming entertainment platforms competing for the same leisure time as live sports

Business / Analysis Opportunity

Since the 2013-14 season, the NBA has captured optical tracking data for every shot taken in every game. That gives us about two million shot attempts with good context — court coordinates, defender distance, shot clock remaining, touch time, dribbles before the shot, and outcome. Combined with player bios, team metadata, and historical performance data going back to 1947, we have a dataset that gives us plenty of features for modeling and visualization and easily clears the project's data requirements.

The opportunity we see is to build an expected shot value model — a way to predict the probability that any given shot goes in based on the context around it — and use that model to separate two things that traditional metrics conflate: shot selection and shot-making skill. Two players with the same field goal percentage can be taking very different shots, and our analysis can quantify that difference.

The chart below shows a representative view of how the league distributes its shots and how often each zone goes in. The pattern frames the opportunity: the shots taken most often sit at opposite ends of the success spectrum, and the mid-range (once a staple of the league) has shrunk dramatically.



Representative example. Actual figures will come from our pulled data during EDA.

Research Questions

We've developed three research questions, each with a clear connection to how the NBA — and the teams that make it up — could use the results.

RQ1 — Which players and teams consistently outperform their expected shot value, and what distinguishes them?

Traditional efficiency metrics like FG% credit players for both shot selection (taking easier shots) and shot-making skill (finishing harder shots). Front offices need to tell the two apart when evaluating talent, signing free agents, and building rosters. By identifying players and teams who consistently beat what our model expects, we can give the NBA and its teams a clearer view of who is creating value beyond what the shots themselves warrant.

RQ2 — How has shot success by court location changed across the 2013-present tracking era?

Over the last decade, the NBA has gone through one of the most dramatic strategic shifts in any major sport — the explosion of three-point shooting. This question asks how the geography of effective shooting has changed, and whether the value of mid-range shots, restricted-area finishes, and corner threes has shifted along with team strategy. The league, broadcast partners, and rules committees all benefit from a quantified view of how the game has evolved.

RQ3 — Does shot-quality-adjusted player ranking differ noticeably from traditional efficiency metrics (FG%, eFG%, TS%)?

If our shot-quality-adjusted metric ranks players very similarly to the metrics already in wide use, then the traditional metrics are doing fine, and the added complexity isn't worth much. If the rankings clearly diverge, then teams currently relying on FG%, eFG%, or TS% may be over-or-under-valuing specific players, with direct implications for contracts, trades, and roster construction.

Hypotheses

For each research question, we have an initial expectation of what we'll find. These are forecasts, not conclusions — we'll let the data confirm or push back.

H1: for RQ1

We expect that players who consistently outperform expected shot value will cluster into a small number of identifiable archetypes; primary shot creators with elite touch, low-usage floor spacers who shoot only when open, and frontcourt players who finish at extremely high rates near the rim. NBA roles are reasonably well-defined, and the players who exceed expectations likely do so for role-specific reasons rather than randomly.

H2: for RQ2

We expect to see a meaningful decline in mid-range shot success over time, alongside an increase in three-point and restricted-area success rates. As teams shifted attempts away from the mid-range, the shots that remain in that zone are likely taken by the league's best mid-range shooters — but the volume drop and devaluation should still show up as reduced league-wide effectiveness from that area.

H3: for RQ3

We expect moderate-to-low rank correlation between our shot-quality-adjusted metric and the traditional metrics, particularly for low-usage role players. Traditional metrics reward players who take easy shots (high-percentage layups, wide-open threes), while a shot-quality-adjusted view should give credit to players who consistently make difficult shots. The two views should disagree most for specialists at the extremes of shot difficulty.

Data

Our analysis draws from one primary source and two supporting datasets. All of them map to NBA.com identifiers, so joins are reliable and reproducible.

Primary Source — NBA Stats API

We're pulling shot detail, game logs, players, and teams from the NBA's official statistics API through the open-source nba_api Python package. The shot detail covers every field goal attempt from the 2013-14 season through the current season with court coordinates, shot zone, defender distance, shot clock remaining, touch time, dribbles before the shot, and the outcome. Game logs, player bios, and team metadata round out the join context.

Supporting Source — Kaggle: NBA Shot Logs 2014-15

A single-season, pre-cleaned dataset of about 128,000 shots from the 2014-15 season, originally scraped from the same NBA Stats API. We're using this for early prototyping while the full multi-season pull from the primary API runs in the background.

Supporting Source — Kaggle: NBA Stats (1947-present)

Maintained by sumitrodatta, this dataset covers NBA, ABA, and BAA stats from 1947 to the present and is actively updated. It gives us long-horizon player history for active players whose careers predate the 2013-14 tracking era — career averages, prior-era context, and career-arc features that the primary API doesn't expose conveniently.

Integration

Every source maps to NBA.com identifiers (player_id, team_id, game_id), so all our joins are ID-based rather than name-based. The product of the data pipeline is one enriched shot-level table — each row is a single shot carrying its own context, the game's context, the shooter's context, and the team's context. That table is what every downstream analysis reads from.

Methodology

RQ1 — Shot-Level Classification, Residual Analysis, and Clustering

RQ1 is layered rather than a single method. The base layer is binary classification at the shot level: we train a model to predict the probability a single shot is made given its context. We'll start with logistic regression as a baseline because of its interpretability, then move to gradient-boosted trees (XGBoost or LightGBM) to capture non-linear interactions between features like defender distance, shot clock, and shot type.

From the trained model, we compute residuals for every player and every team — actual makes minus model-expected makes. Players with consistently positive residuals are our overperformers. To answer the 'what distinguishes them' part of the question, we cluster those overperformers using playing-style features (usage rate, shot mix, position, team pace) to identify archetypes.

RQ2 — Spatial Binning, Smoothed Heat Maps, and Longitudinal Comparison

RQ2 is descriptive rather than predictive. We'll bin the court into standard NBA shot zones (restricted area, in-the-paint non-RA, mid-range, corner three, above-the-break three), compute shot success rate per zone for each season from 2013-14 through the present and visualize using kernel density estimation to produce smooth heat maps. To quantify whether changes between seasons are statistically meaningful, we'll use paired comparisons of zone-level shooting percentages across eras and potentially fit a model that includes season as a feature to estimate how location's effect on success has shifted over time.

RQ3 — Rank Correlation and Visualization

We'll build a shot-quality-adjusted player metric directly from the RQ1 model — something like expected FG% above league average. We'll then compute the traditional metrics (FG%, eFG%, TS%) on the same dataset for an apples-to-apples comparison. Spearman's rho will quantify how similar the two rankings are, and a slope chart will visualize which specific players move up or down the most, under the adjusted metric. The interpretation focuses on whether certain player types or roles systematically get re-ranked.

Computational Methods and Outputs

The specific techniques we expect to use:

- Logistic regression (baseline classifier for shot success)
- Gradient-boosted trees, likely XGBoost or LightGBM (main shot success model)
- Residual analysis (player and team aggregations of model error)
- K-means or hierarchical clustering (player archetype identification for RQ1)
- Kernel density estimation (smoothed shot success heat maps for RQ2)
- Paired statistical tests (longitudinal comparison for RQ2)
- Spearman rank correlation (RQ3)
- Slope charts and rank-shift visualizations (RQ3)

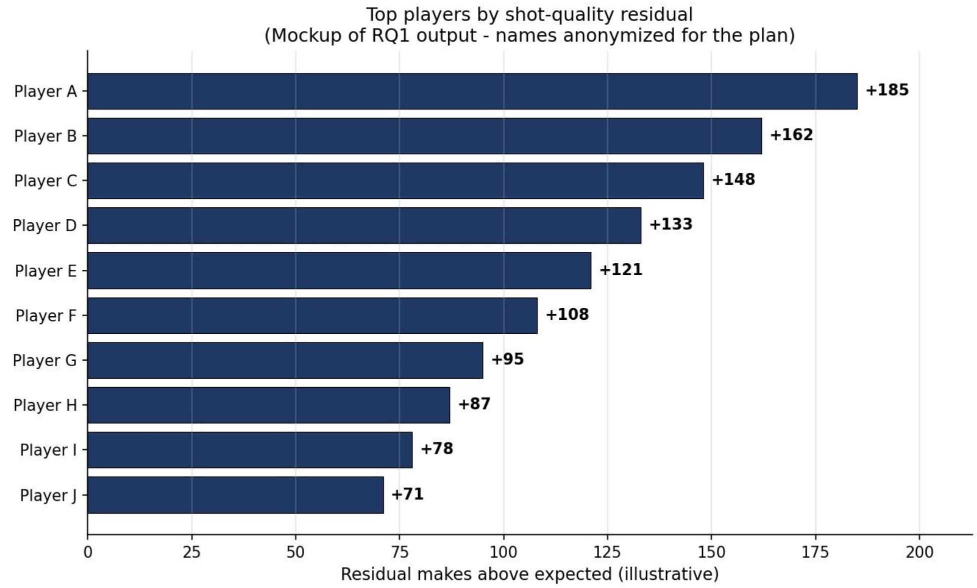
All source code for the pipeline, modeling, and visualizations will be included in the Appendix of the Final Report rather than in the body.

Output Mockups

These are early-stage mockups showing the shape our final outputs will take. The actual outputs will be generated from our pulled data once the pipeline is in place.

RQ1 — Overperformer Residual Leaderboard and Archetypes

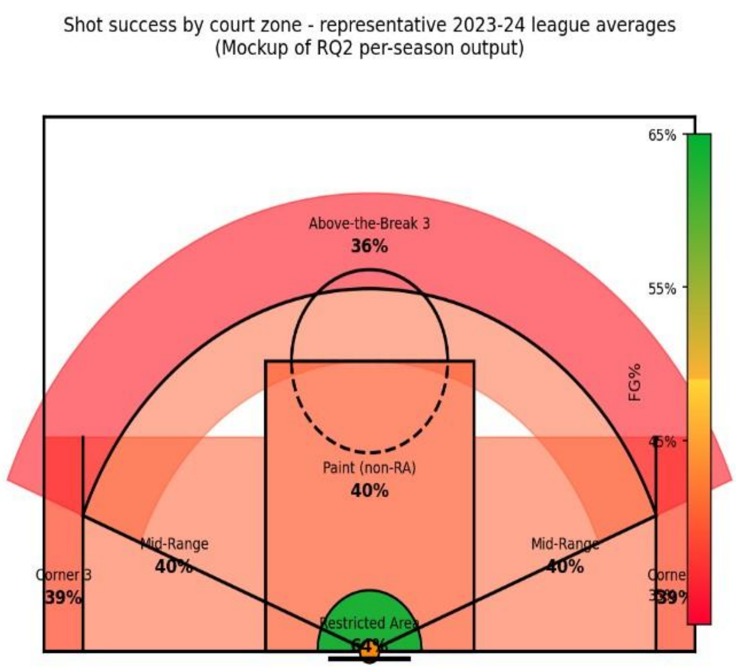
Our primary RQ1 output is a leaderboard of the top players (and separately, top teams) by total residual shot value over the tracking era. Each row shows the player's actual makes, expected makes from the model, and the difference. A secondary output describes the player archetypes that emerge from clustering; for example, 'elite primary creators,' 'high-volume catch-and-shoot specialists,' and 'efficient interior finishers.'



Mockup of the RQ1 leaderboard. Player names are anonymized for the plan; real results will name actual players from our analysis.

RQ2 — Heat Maps Across the Tracking Era

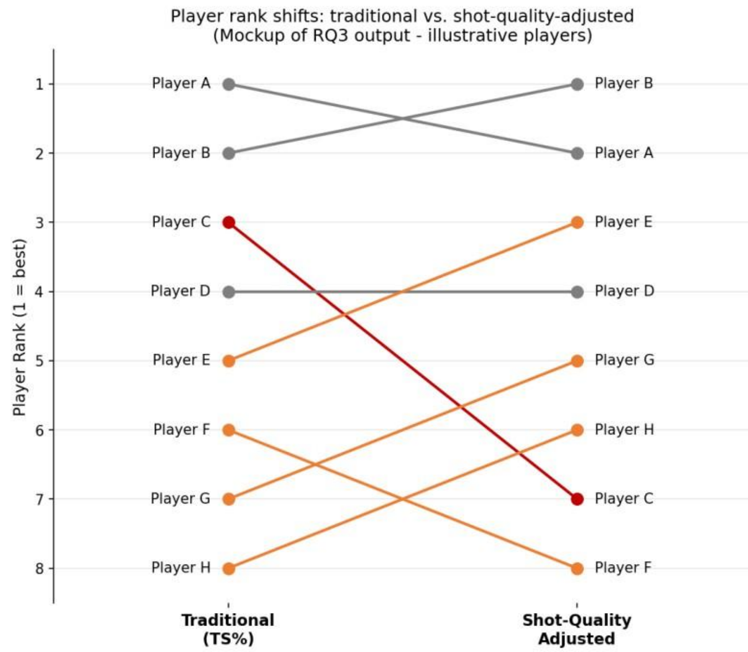
Our RQ2 output is a series of smoothed shot-success heat maps, one per season, that can be viewed side by side to show how the geography of effective shooting has changed. A summary chart shows zone-level shooting percentages plotted over time.



Mockup of the per-season heat map output. Real heat maps will be generated per season from our shot detail data.

RQ3 — Rank-Shift Slope Chart

The RQ3 output is a slope chart with two parallel rank axes — players ranked by traditional efficiency metrics on the left, ranked by our shot-quality-adjusted metric on the right — with lines connecting each player's two ranks. Large slopes flag the players whose evaluation changes most under the adjusted metric. We'll also report the Spearman rank correlation as a single summary statistic.



Mockup of the RQ3 slope chart. Real player names and rankings will replace the placeholders.

Campaign Implementation

Once the analysis is done, the results have clear applications for several groups in and around the NBA.

Team front offices and basketball operations staff are the most direct audience. The shot-quality-adjusted player rankings from RQ3 give general managers a sharper tool for free-agent evaluation, contract negotiation, and trade analysis. The archetype clusters from RQ1 help inform roster construction — knowing which kinds of players consistently exceed their expected value lets teams target specific profiles in the draft and trade market.

The NBA league office is interested in the league-wide trends surfaced by RQ2. The continued evolution of shot selection has implications for rules changes, court dimensions, and how the game is officiated and presented. Quantifying how location's value has shifted gives the league empirical grounding for decisions about the game's strategic direction.

Broadcast partners (Disney/ABC/ESPN, NBCUniversal, Amazon Prime Video) benefit from richer storytelling around individual player performance. A broadcast graphic showing that a player is among the league leaders in shot-quality-adjusted scoring tells a more meaningful story than raw shooting percentages, especially for role players whose contributions get overlooked in traditional box scores.

Player agents and media analysts can use the same metrics to advocate for their clients or surface narratives traditional stats miss — particularly for low-usage specialists whose value gets undersold by conventional efficiency measures.

Timeline & Milestones

Below is our planned schedule for the rest of the semester. This is an estimate, as the project plan is a living document, and we'll communicate any significant schedule shifts.

Week	Dates	Milestones
Week 2 *(current)*	**May 25 – 31**	Project Plan submitted; Research Questions submitted; repo and data pipeline scaffolded; Germain begins running the full NBA Stats API pulls; Cole drafts and submits the Literature Review.
Week 3	**June 1 – 7**	Data pulls complete and enriched shot table built; Calder runs EDA on the shot-level data; Marc prototypes the shot-success model on the 2014–15 Kaggle dataset; Cole drafts the Methodology and EDA writeups, submitted as each is finalized.
Week 4	**June 8 – 14**	Marc trains the full multi-season model and runs residual analysis + clustering for RQ1; RQ2 heat-map series and RQ3 rank comparison generated; Cole drafts the Ethics and Data Visualizations writeups, submitted as each is finalized.
Week 5	**June 15 – 21**	Analysis, Challenges, Recommendations & Next Steps, Abstract, and References drafted; Rough Draft compiled and reviewed by the team. **Rough Draft due Sunday, June 21 at 11:59 PM.**
Week 6	**June 22 – 26**	Instructor feedback received on the Rough Draft; revisions incorporated within the 50% grade-recovery window; source code Appendix finalized; full document polished and proofread. **Final Report due Friday, June 26 at 11:59 PM.**

References

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This plan is structured to match the components in the official DAT 490 Summer 2026 assignment spec.